

Running the a1954 ^{12}Be analysis

Step-by-step guide to the reconstruction, pattern-recognition and track-fitting pipeline, with the tuning knobs that matter

ATTPCROOT 0openKF-Claude branch

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Abstract

This guide walks through the complete analysis of the a1954 ^{12}Be proton-target data set — $^{12}\text{Be}(p, p')^{12}\text{Be}^*$ and $^{12}\text{Be}(p, d)^{11}\text{Be}$ in inverse kinematics — from the raw HDF5 files to the excitation-energy spectrum. Every stage is given as a command you can copy, followed by the options that stage exposes. Sections 9 and 10 collect the pattern-recognition (PRA) and fitting parameters in reference tables together with recipes for improving them, and Section 11 lists the traps this data set has already sprung.

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1 What the analysis does

A ^{12}Be beam at ≈ 155 MeV (12.9 MeV/u, *not yet independently confirmed*; see Section 11) is stopped in the AT-TPC filled with H_2 at 600 torr, which acts simultaneously as target and detection medium. Two channels are analysed:

Channel	Ejectile detected	Typical θ_{lab}	Typical KE	Residual
$^{12}\text{Be}(p, p')^{12}\text{Be}^*$	recoil proton	60–90° (backward)	1–10 MeV	^{12}Be
$^{12}\text{Be}(p, d)^{11}\text{Be}$	deuteron	10–50° (forward)	10–30 MeV	^{11}Be

The pipeline is staged so that the expensive part (unpacking + PSA + PRA, hours per run) is done *once* and the cheap part (fitting, minutes per run) can be iterated freely:

```
run_XXXX.h5 → reco (unpack + PSA + clean + PRA) → run_XXXX_reco.root → slim → gate (IC +
PID) → fit (UKF and/or GENFIT) → kinematics ( $E_x$  spectrum, explorers)
```

All macros live in macro/Unpack_HDF5/a1954_Be12/UKF/ of the OpenKF-Claude branch, referred to below as \$HERE.

2 Setting up

2.1 Environment

```
export REPO=/home/yassid/fair_install/ATTPCR00Tv2-OpenKF
export HERE=$REPO/macro/Unpack_HDF5/a1954_Be12/UKF

source /home/yassid/fair_install/FairSoft/install/bin/thisroot.sh
source $REPO/build/config.sh
# MANDATORY -- config.sh does not set this, and cling crashes without it
export ROOT_INCLUDE_PATH=$REPO/build/include:/home/yassid/fair_install/FairRootInstall/include
```

ROOT_INCLUDE_PATH must point at build/include, not at the source directories. Pointing it at the sources makes cling crash on a hidden-friend `swap` declaration. Every batch script in \$HERE already exports it correctly; you only need this line for interactive work.

2.2 Two builds, on purpose

Build	GENFIT2	Use for
build/	not found	reco, PSA, PRA, UKF fitting
build_genfit/	linked against <code>~/fair_install/GenFitInst</code>	<code>fitGenfit_Be12.C</code>

They compile the same sources; they are kept apart so a GENFIT rebuild never disturbs a running reco batch. Source `build_genfit/config.sh` instead of `build/config.sh` when you run GENFIT.

2.3 Geometry (once per clone)

The geometry `.root` files are *not* tracked in git. Build the 600 torr hydrogen geometry that GENFIT needs before the first GENFIT fit:

```
cd $REPO && root -b -q geometry/ATTPC_H600torr.C
# writes geometry/ATTPC_H600torr.root and ATTPC_H600torr_geomanager.root (6.616e-5 g/cm3)
```

Do *not* fall back on `ATTPC_H1bar_geomanager.root`: a1954 ran at 600 torr, so the 1-bar geometry is 27% too dense and every material-effects correction GENFIT applies will be miscalibrated.

2.4 Data

Raw HDF5	<code>/mnt/f/a1954_remerged/run_XXXX.h5</code> (drive F, WD_Cris)
Mount	<code>sudo mount -t drvfs F: /mnt/f</code> — F: is <i>not</i> auto-mounted after a reboot
Physics runs	0142–0215 (+0216–0224); 0160 and 0177–0181, 0183 are absent
Runs fitted	75 (<code>~/a1954_Be12_fit/</code>); 73 clean after dropping the two desynced runs
Excluded	run_0142 and run_0212 (timestamp desync, Section 5.2); run_0161 in the (<i>p</i> , <i>d</i>) pass (FRIB file empty, 587 B)
Parameters	<code>parameters/ATTPC.a1954_Be12.par</code>

Key entries of the parameter file:

Parameter	Value	Comment
BField	2.85 T	sign is set in the macros, <i>not</i> here
GasPressure	600 torr	H ₂ , 293 K, 6.616e-5 g/cm ³
DriftVelocity	1.00 cm/ μs	<i>the</i> parameter to scan (Section 10.4)
SamplingRate	3.0	MHz
TBEntrance	300	beam entrance in time buckets
NumTbs	512	
ZPadPlane	1000 mm	
PeakingTime	720 ns	GET electronics τ

The ^{14}C sibling analysis (`macro/Unpack_HDF5/a1954/UKF/`) uses a *different* par file with `SamplingRate 6.0`, `DriftVelocity 1.30` and `TBEntrance 480`. Do not mix them up.

3 Stage 1 — reconstruction (unpack + PSA + cleaning + PRA)

This is the expensive stage: it reads the 5–9 GB HDF5 file, turns pad traces into hits, removes noise hits and clusters the surviving hits into track candidates.

3.1 One run, interactively

```
root -b -q -l "$HERE/pipeline/unpackReco_Be12.C(\"run_0147\")"
# first 500 events only, for a quick look:
root -b -q -l "$HERE/pipeline/unpackReco_Be12.C(\"run_0147\",500)"
```

3.2 Production (the validated configuration)

```
cd $HERE
./reco_hdb_F.sh "run_0142 run_0143 run_0144" 2      # 2-parallel; drufs I/O is the cap
```

reco_hdb_F.sh reads from /mnt/f/a1954_remerged/ and writes /mnt/f/a1954_Be12_reco_hdb/run_XXXX_reco.root. It skips runs that are already done and logs to \$OUT/logs/. The configuration it pins is:

AtPSAMultiFit (thr 20) + AtDirDeDxCleaner + AtTrackFinderHDBSCAN (mover join)

3.3 Full argument list of unpackReco_Be12.C

#	Argument	Default	Meaning
1	fileName	"run_0142"	run name, no extension
2	nEvents	0	0 = all events
3	persistRaw	true	write the raw pads too; production uses false (files are 4× smaller)
4	outDir	~/a1954_Be12_reco/	output directory
5	filepath	(NSCL drive)	input directory; use /mnt/f/a1954_remerged/
6	doSC	false	space-charge correction; not needed (does not affect z)
7	applyTimeCorr	false	pad-time offsets — the CSV that exists is a1975-specific; leave OFF until an a1954 table is measured
8	psaType	"multifit"	"multifit" or "max"
9	primSigma	0	> 0 enables prominence-on-primary at that many σ
10	thr	20	PSA amplitude threshold [ADC]
11	praType	"tc"	"tc" (triplclust) or "hdbscan" — production uses hdbscan
12	hdMcs	20	HDBSCAN min cluster size
13	hdMs	8	HDBSCAN min samples
14	fitChi2	0	> 0 = PSA shape χ^2 gate (also turns on float- τ); < 0 = diagnostic only
15	relErr	0.1	fractional model error in the PSA χ^2
16	parFile	ATTPC.a1954_Be12.par	
17	doClean	kTRUE	run AtDirDeDxCleaner before PRA

3.4 What each sub-stage produces

Task	Branch out	Content
AtUnpackTask	AtRawEvent	raw pad traces (only if <code>persistRaw</code>)
AtPSATask	AtEventH	AtHit point cloud
AtDataCleaningTask	AtEventClean	de-noised point cloud (transient)
AtPRATask	AtPatternEvent	AtTrack candidates + their hits

4 Stage 2 — FRIB/ion-chamber unpacking and the slim cache

The ion chamber lives in the `frib/evt` HDF5 group, not in the pad stream, and is unpacked separately by `AtFRIBHDFUnpacker`. It is fast (no PSA/PRA).

```
cd $HERE
./frib_F_batch.sh "run_0142 run_0143 ..." 4 # -> ~/a1954_Be12_reco_hdb_slim/run_XXX_FRIB.root
./slim_cache_batch.sh "run_0142 run_0143 ..." 4 # -> ~/a1954_Be12_reco_hdb_slim/run_XXX_slim.
↪ root
```

The *slim cache* clones only the `AtPatternEvent` and `EventHeader` branches of the big reco file onto the local disk. Every downstream PID/gate/fit pass then reads locally instead of streaming tens of GB back over drvfs ($\approx 250\times$ faster).

An empty FRIB file (a few hundred bytes, e.g. `run_0161`) means no IC gate is possible for that run. The batch scripts skip files below 10 kB.

5 Stage 3 — beam identification and particle ID

5.1 The ion-chamber gate

The a1954 beam is a fragment cocktail. The IC pulse is *bipolar*: a positive overshoot near TB 1134 and a negative dip near TB 1155.

IC observable	maximum ADC in TB [1050, 1250] (the positive overshoot, <i>not</i> $-\min$)
¹² Be	the isolated peak at $\approx 650 \Rightarrow$ gate [500, 900]
Contaminant	the dominant peak at ≈ 1900 — reject
Pile-up	require exactly one pulse above 200 ADC in TB [800, 1500]

```
root -b -q -l "$HERE/pid/ic_Be12.C(\"run_0147,run_0148\", \"~/a1954_Be12_reco_hdb_slim/\")"
root -b -q -l "$HERE/pid/ic_multiplicity_Be12.C(\"run_0147\")" # pile-up check
```

5.2 Timestamp synchronisation — run this before trusting any gate

The IC is matched to the pad data **by entry index**, which is only valid when the two streams have not dropped events relative to each other. The check compares timestamp *deltas* (so it is offset-independent):

```
root -b -q -l "$HERE/pid/tsync_check.C(\"run_0143\")"
```

Verdict	Signature	Action
aligned	event-count difference +1 (trailing event), $\approx 100\%$ deltas agree	use
desynced	negative count difference, 0% agree	exclude the run

For a1954 ¹²Be: 73 of 75 runs are aligned; **run_0142** (diff -9) and **run_0212** (diff -5) are desynced and must be dropped.

5.3 Drawing the PID gate

The PID plane is $\sqrt{dE/dx}$ versus $B\rho$ from `AtTools::AtSpyralPID`, computed on the PRA track candidates — *no fit is needed*, so this is the natural place to inspect the data before committing to a fitting configuration. It is also charge-sign agnostic, hence unaffected by the B_z handedness.

```
# 1. build the PID plane and persist it as a CSV
root -b -q -l "$HERE/pid/pid_Be12.C(\"run_0147,run_0148\", \"~/a1954_Be12_reco_hdb_slim/\")"
root -b -q -l "$HERE/pid/dump_pid_Be12.C(\"run_0147\", \"<recoDir>\", \"$HERE/pid/pid_Be12_data.csv
↪ \")"

# 2. draw the polygon by hand (INTERACTIVE -- never with -b)
root -l "$HERE/pid/draw_gate_Be12.C(\"proton_12Be\")"
root -l "$HERE/pid/draw_gate_Be12.C(\"deuteron_12Be\", 625, 750, 0, 1, 2)"

# 3. verify it headlessly
root -b -q -l "$HERE/pid/apply_gate_Be12.C(\"proton_12Be.json\")"
root -b -q -l "$HERE/pid/show_gates_Be12.C"
```

Left-click to place each vertex, double-click to close; the polygon is written to `<name>.json` in `$HERE/pid/`. The gates in use are `pid/proton_12Be.json` (wraps the full proton band) and `pid/deuteron_12Be.json`.

An `arclen > 200` cut visibly sharpens the band and is worth trying when redrawing a gate. HDBSCAN reco gives a tighter band than triplclust.

6 Stage 4 — building the gated fit input

`gate_events_Be12.C` writes a small ROOT file containing *only* the events that pass the IC gate and, inside them, *only* the tracks that fall in the PID polygon and the requested angular window. Because it also copies the FairRoot metadata (`BranchList`, `FileHeader`, `cbmout`), the fitters read it exactly like a normal reco file.

```
root -b -q -l "$HERE/pipeline/gate_events_Be12.C(\"run_0147\", \"
  \"/home/yassid/a1954_Be12_reco_hdb_slim/\", \"~/home/yassid/a1954_Be12_fit/in/\", \"
  \"/mnt/f/a1954_Be12_reco_hdb/run_0147_reco.root\")"
```

#	Argument	Default	Meaning
1	<code>run</code>	—	run name
2	<code>inDir</code>	<code>slim dir</code>	holds <code>_slim.root</code> and <code>_FRIB.root</code>
3	<code>outDir</code>	<code>~/a1954_Be12_fit/in/</code>	gated output
4	<code>refReco</code>	<code>""</code>	big reco file, only to copy FairRoot meta-data from
5,6	<code>icLo, icHi</code>	<code>500, 900</code>	IC amplitude window
7,8	<code>icTbLo, icTbHi</code>	<code>1050, 1250</code>	IC integration window [TB]
9	<code>bField</code>	<code>2.85</code>	used by <code>AtSpyralPID</code> (magnitude only)
10	<code>protonGate</code>	<code>pid/proton_12Be.json</code>	PID polygon; swap for <code>deuteron_12Be.json</code>
11	<code>thMin</code>	<code>90.0</code>	minimum θ_{lab} [deg]. 90 for (p, p'); 0 for (p, d)
12	<code>peakThr</code>	<code>200</code>	IC pulse-counting threshold
13,14	<code>pkTbLo, pkTbHi</code>	<code>800, 1500</code>	IC pulse-counting window

The θ convention differs between stages. The `thMin` cut here is on the *Spyral* polar angle `r.polar`, whereas the fitted `k.theta` used downstream satisfies $\theta_{\text{fit}} \approx 180^\circ - \theta_{\text{polar}}$. Using `thMin = 90` on a (p, d) pass kills every deuteron.

7 Stage 5 — fitting

Both fitters read the *same* gated input, so tracks can be compared one-to-one.

7.1 The (p, p') production pass

```
cd $HERE
./fitpipe_Be12.sh "run_0143 run_0144 run_0145" 4      # gate -> UKF -> GENFIT, 4 cores
# outputs: ~/a1954_Be12_fit/run_XXX_{ukf,genfit}.root (+ rsync to /mnt/f/a1954_Be12_fit/)
```

7.2 The (p, d) production pass

```
./fitpipe_pd_Be12.sh "run_0143 run_0144" 4 0      # deuteron gate, thMin = 0 (FORWARD tracks)
./genfit_pd_Be12.sh "run_0143 run_0144" 4      # GENFIT over the same gated input
# outputs: ~/a1954_Be12_fit_pd/ and ~/a1954_Be12_genfix_pd/
```

7.3 UKF — fitUKF_Be12.C

```
root -b -q -l "$HERE/pipeline/fitUKF_Be12.C(\"run_0147\", -1, \"proton\", -1, 2.85, 6.616e-5, \
  \"\", \"/home/yassid/a1954_Be12_fit_600torr/in/\", 0.5, 0.1, 1, 10, \"/home/yassid/a1954_Be12_fit_600
  ↪ torr/\")"
```

#	Argument	Default	Meaning
1	fileName	"run_0142"	
2	nEvents	-1	-1 = all
3	particles	"proton"	comma list; one hypothesis per species (proton, deuteron, triton, He3, alpha)
4	bFieldSign	-1	-1 = experimental handedness , +1 = simulation
5	bFieldMag	2.85	T
6	gasDensity	3.308e-5	g/cm ³ , feeds CATIMA. Pass 6.616e-5 (H ₂ 600 torr); the default is the wrong 300 torr
7	outSuffix	""	appended to the output name (use for scans)
8	ioDir	reco dir	input directory
9	measSigma	0.5	per-cluster position σ [mm]
10	momSigmaFrac	0.1	fractional momentum-seed uncertainty
11	nIter	1	filter/smoothen iterations
12	minClusters	10	minimum clusters for a fit attempt
13	outDir	""	defaults to ioDir
14	refTrack	false	reference-track mode
15	refInflation	4.0	covariance inflation in reference-track mode

Handedness. Experimental data has the opposite helix handedness to simulation because the digitisation reverses the drift z . Use $B_z = -2.85$ T for data. This was validated on run_0142: median $\chi^2/\text{ndf} = 0.03$, median KE 5.96 MeV, clean elastic locus. If fits diverge or the KE is unphysical, flip the sign first.

7.4 GENFIT — fitGenfit_Be12.C

```
source $REPO/build_genfit/config.sh          # <-- the GENFIT build
root -b -q -l "$HERE/pipeline/fitGenfit_Be12.C(\"run_0147\",-1,\"<inDir>\",\"\", \"<outDir>\",\
-2.85,2,5,\"\",4.0,10.0,170.0,kTRUE,kFALSE,\"proton\", \"ATTPC_H600torr\")"
```

#	Argument	Default	Meaning
1	fileName	"run_0142"	
2	nEvents	-1	
3	ioDir	reco dir	
4	outSuffix	" "	
5	outDir	" "	
6	bField	-2.85	signed, as for the UKF
7,8	minIter, maxIter	2, 5	Kalman refit iterations
9	pidGate	" "	optional in-fitter Spyral PID gate (JSON)
10	measSigma	4.0	per-cluster position σ [mm]
11,12	thetaMinDeg, thetaMaxDeg	10, 170	lower it to 5 for (p, d)
13	matEffects	kTRUE	keep it — see below
14	backwardSeedFix	kFALSE	alternative backward-track seeding
15	particle	"proton"	proton, deuteron, triton
16	geoName	ATTPC_H300torr_RT	must match the real gas density: pass ATTPC_H600torr; the default is the wrong 300 torr

Always run GENFIT with matEffects = kTRUE and the 600-torr geometry. With matEffects = kFALSE GENFIT fits a constant-momentum helix, so the reported KE is a track *average*: it comes out $\approx 30\%$ low below 6 MeV, 9.5% low at 6–10 MeV, and only converges to the UKF above ≈ 10 –15 MeV. That single flag was the whole +0.45 MeV E_x offset.

7.5 Which fitter to believe

After the `AtGenfitter` energy-loss fix (commit 5f87e625) the two fitters agree. The bug: the code called `useEnergyLossParam()` unconditionally, which threw on every step and silently truncated reference tracks to $\text{ndf} \leq 0$.

(p, p') , 73 clean runs, $\chi^2/\text{ndf} < 5$	tracks	g.s. μ [MeV]	FWHM [MeV]
UKF	26 604	+0.020	0.555
GENFIT, pre-fix (matEffects=kFALSE)	12 533	+0.451	—
GENFIT, fixed	23 316	+0.028	0.521

GENFIT is now marginally the better resolution and its old KE-vs- θ double band is gone. A known residual remains: a +16% KE deficit *below 2 MeV*, because under $\beta\gamma = 0.05$ genfit leaves Bethe–Bloch for `dEdxParam`, which returns zero without a table. Load a proton dE/dx table via `SetELossFile()` to close it.

For (p, d) : UKF 2661 vs GENFIT 2866 deuterons; both behave, because forward deuterons are 10–30 MeV where GENFIT was never in trouble.

8 Stage 6 — physics output

8.1 Excitation-energy spectrum

`ex_Be12.C` applies relativistic two-body kinematics to the fitted (KE, θ) and fills the E_x spectrum, E_x vs θ_{cm} , and KE vs θ_{lab} . It also writes an `TNtuple` cache (`ke:theta:vertexz:thcm:ex:chi2ndf`) that every interactive tool then works from.

```
RUNS="run_0143,run_0144,run_0145"    # comma separated!

# (p,p') -- proton ejectile, 12Be residual
root -b -q -l "$HERE/pp/ex_Be12.C(\"$RUNS\", \" /home/yassid/a1954_Be12_genfix/\", 155, 5, \"_pp_genfit\", 1.007825, 12.026921, \"12Be(p,p')\", \"genfit\")"

# (p,d) -- deuteron ejectile, 11Be residual
root -b -q -l "$HERE/pp/ex_Be12.C(\"$RUNS\", \" /home/yassid/a1954_Be12_fit_pd/\", 155, 5, \"_pd_ukf\", 2.014102, 11.021658, \"12Be(p,d)11Be\", \"ukf\")"
```

#	Argument	Meaning
1	runsCSV	comma-separated run list
2	inDir	directory holding <code><run>_<fitter>.root</code>
3	Ebeam	beam energy [MeV] — 155 for a1954 ^{12}Be
4	chi2Cut	χ^2/ndf cut (5 nominal; 1e9 to disable)
5	outTag	cache suffix — must include the leading underscore
6	mEjectAmu	1.007825 (p) or 2.014102 (d)
7	mResidAmu	12.026921 (^{12}Be) or 11.021658 (^{11}Be)
8	chanLabel	plot title
9	fitter	"ukf" or "genfit" — selects the input file suffix

Caches land in `$HERE/pp/plots/proton_kin<outTag>.root`.

8.2 Interactive exploration

Both explorers recompute E_x and θ_{cm} live from the cache, so you can move E_{beam} , the cuts and the binnings without refitting anything. Both carry the UKF and GENFIT caches with an in-page switch and an overlay.

```
# browser version (no X11 needed; regenerates the HTML and launches Chrome)
$HERE/pp/open_explorer.sh pp          # 12Be(p,p')
$HERE/pp/open_explorer.sh pd          # 12Be(p,d)11Be

# ROOT/X11 GUI version -- plain ROOT, no config.sh needed, and never with -b
root -l $HERE/pp/explore_Be12.C
```

ROOT's 3D/Eve viewers crash under WSL, and `explorer.exe <file>` silently does nothing on this machine — `open_explorer.sh` therefore stages the HTML on the Windows side and launches the browser binary directly. Prefer the browser explorer.

8.3 Fitter comparison

```
root -b -q -l "$HERE/pp/compare_fitters_Be12.C(\"run_0143,run_0144\", \" /home/yassid/a1954_Be12_fit/\")"
# -> pp/plots/compare_fitters_Be12.png, compare_fitters_chi2_Be12.png
```

9 Tuning the pattern recognition

PRA and fitting settings are physics-dependent. A knob that improves one reaction/geometry can hurt another. Always re-validate on *this* data set — one run, a few hundred events, is enough to see the effect.

9.1 Choosing the track finder

	AtTrackFinderTC (triplclust)	AtTrackFinderHDBSCAN
Selected by	<code>praType="tc"</code>	<code>praType="hdbscan"</code>
Strength	fast, robust on isolated tracks	separates <i>overlapping/crossing</i> tracks that triplclust merges
Weakness	merges crossing tracks	over-segments; needs the join step
On a1954	tighter-band alternative	production choice ; gives the tightest PID band

9.2 HDBSCAN parameters

Set inside `unpackReco_Be12.C`; the first two are exposed as macro arguments (`hdMcs`, `hdMs`), the rest are edited in the macro body.

Setter	a1954 value	Default	Effect / how to tune
<code>SetMinClusterSize</code>	20	adaptive	Smallest accepted cluster. Lower to recover short/low-energy tracks, raise to suppress noise blobs. Primary yield knob.
<code>SetMinSamples</code>	8	3	k for the core distance. Higher = more conservative, more points declared noise.
<code>SetClusterSelectionEpsilon</code>	10.0	10.0	[mm] Do not split clusters closer than this. The big over-segmentation lever : raise to stop one track being cut into pieces, lower to separate two nearby tracks.
<code>SetJoinMethod</code>	"mover"	"overlap"	<code>mover</code> = vertex-seeded trajectory following; <code>overlap</code> = circle-area overlap; <code>continuity</code> = end-to-end; <code>none</code> .
<code>SetMinClusterSizeJoin</code>	15	15	Minimum size a fragment must have to take part in a join.
<code>SetCircleOverlapRatio</code>	0.25	0.25	Overlap join: merge if overlap area $> \text{ratio} \times \text{smaller area}$.
<code>SetMotionGapTol</code>	40	40	[mm] <code>mover</code> : largest end-to-end gap that can be chained. Raise if tracks stay broken across a gap.
<code>SetMotionAngleTol</code>	35	35	[deg] <code>mover</code> : tangent-continuity tolerance at the junction.
<code>SetOutlierScaleFactor</code>	— (0.05)	0.05	Per-cluster LOF cleaning; ≤ 0 disables.
<code>SetOutlierThreshold</code>	— (1.5)	1.5	LOF above this $\rightarrow$ noise.
<code>SetJoinZFraction</code>	— (0.2)	0.2	<code>continuity</code> join only.

Setter	a1954 value	Default	Effect / how to tune
<code>SetJoinRadiusFraction</code>	— (0.3)	0.3	continuity join only.

9.3 Triplclust parameters

Exposed through `SetScluster` (s , 0.3), `SetKtriplet` (k , 19), `SetNtriplet` (n , 2), `SetMcluster` (m , 15), `SetRsmooth` (r , 2), `SetAtriplet` (a , 0.03), `SetTcluster` (t , 4.0); a1954 additionally sets the base-class `SetClusterRadius`(15.0) and `SetClusterDistance`(7.5). Increasing k and m makes clustering more conservative (fewer, cleaner tracks); increasing s merges more aggressively.

9.4 Base-class knobs shared by both finders (AtPRA)

Setter	Default	Effect
<code>SetMinHits</code> / <code>SetMaxHits</code>	0 / 5000	accepted hit-count window per track
<code>SetClusterRadius</code>	0 (15 for tc)	[mm] radius of the hit clusters handed to the fitter
<code>SetClusterDistance</code>	0 (7.5 for tc)	[mm] spacing between those clusters
<code>SetTargetClusters</code>	25	adaptive clustering target
<code>SetUseArcWalk</code>	false	arc-walk clustering instead of <code>ClusterizeSmooth3D</code> ; recovers track tails
<code>SetUseDriftAwareWeights</code>	false	weight the circle fit by $1/\sigma_{xy}^2(z)$
<code>SetMergeAngleThreshold</code>	45 deg	fragment-merging angular tolerance

The cluster radius/distance pair is the hidden coupling between PRA and fitting: it sets how many measurement points the Kalman filter sees. Too few clusters and the fit is under-constrained (ndf collapses); too many and you are fitting correlated noise. Change it and re-check `minClusters` in the fitter.

9.5 PSA parameters worth revisiting

Setter	a1954	Effect
<code>SetThreshold</code>	20	ADC threshold. The single biggest hit-yield knob; lower it to recover faint tracks at the cost of noise (the cleaner then has more work).
<code>SetPeakingTime</code>	0.720	[μs] GET τ ; must match the electronics.
<code>SetMaxPeaks</code>	4	pulses fitted simultaneously per pad — this is what deconvolves two tracks sharing a pad.
<code>SetMinSeparation</code>	4	[TB] minimum seed spacing; lower resolves closer crossings.
<code>SetProminenceOnPrimary</code>	false	leave false. True rejects broad baseline humps but also kills diffusion-broadened far-drift pulses, collapsing the track in z .
<code>SetFitChi2Cut</code>	0 (off)	shape gate on the fitted pulse; also enables float- τ .
<code>SetMinSignificance</code>	3.0	keep a pulse only if $A/\sigma_A \geq$ this.

9.6 Cleaner parameters (AtDirDeDxCleaner)

Constructor defaults: $k = 12$, `cosSeg` 0.78, `cosTan` 0.72, `rMax` 32 mm, `qRatio` 0.65, `minDeg` 1, `smoothQ` true, `qKeep` 150. A hit survives if it has at least `minDeg` neighbours that are both direction- and dE/dx -continuous. Relax `cosSeg/cosTan` or raise `rMax` if the cleaner is eating sparse low- dE/dx tracks; raise `minDeg` if noise is surviving.

9.7 A practical PRA tuning loop

1. Pick one representative run and reco only 500–2000 events into a scratch directory, varying *one* knob:

```
root -b -q 'pipeline/unpackReco_Be12.C("run_0147",2000,false,"/tmp/scan_mcs15/",...)'
```

2. Judge it on the **PID plane** first (`pid/pid_Be12.C`) — it needs no fit and a tighter, more continuous band is the cleanest indicator that clustering improved.
3. Then look at track multiplicity per event and at the fraction of events with at least one gated track (printed by `gate_events_Be12.C`).
4. Only then fit, and judge on the elastic peak: **FWHM of the E_x g.s. peak** and the number of surviving tracks. Both are printed by `ex_Be12.C` and fitted live in the explorers.
5. Keep the scan outputs under distinct `outSuffix/outTag` names so caches never overwrite each other.

10 Tuning the fitters

10.1 measSigma — and why the χ^2 cut is not portable

The UKF runs at `measSigma` = 0.5 mm and GENFIT at 4.0 mm. Both enter identically as a per-cluster position sigma, so the $8\times$ difference is $64\times$ in χ^2 . This was measured directly on 73 runs by swapping the two values (`sigma_test_Be12.sh`):

	tracks (no χ^2 cut)	kept by $\chi^2/\text{ndf} < 5$
UKF at $\sigma = 0.5$	32 120	82.8 %
UKF at $\sigma = 4.0$	—	99.5 %
GENFIT at $\sigma = 4.0$	31 578	39.7 %
GENFIT at $\sigma = 0.5$	—	4.2 %

Swapping the sigmas scaled GENFIT’s median χ^2 by 63.9 (predicted $(4/0.5)^2 = 64$) and the UKF’s by 40.1 — less than 64, because the UKF’s `measSigma` also enters the filter gain.

A χ^2/ndf cut is not portable between fitters or between `measSigma` settings. With no cut, both fitters return the same tracks to within 1.7%; the shared cut at 5 is the *entire* 26 604 vs 12 533 yield gap. Set the cut per fitter from a percentile of that fitter’s own χ^2 distribution, not from a number copied across.

Importantly, `measSigma` does *not* move the energy scale: the GENFIT g.s. centroid was +0.451 MeV at both $\sigma = 4.0$ and $\sigma = 0.5$ on 31.5 k tracks each. Covariance controls *which tracks survive a cut*, not *what energy they get*.

10.2 UKF knobs

Knob	a1954	Effect / when to change
<code>measSigma</code>	0.5 mm	See above. Raise towards the physical cluster resolution if you want a χ^2 that is interpretable; then re-derive the cut.
<code>momSigmaFrac</code>	0.1	Momentum-seed uncertainty. Raise if the seed is poor (short tracks) so the filter is not over-confident.
<code>nIter</code>	1	More iterations help when the seed is far from the solution; costs time linearly.

Knob	a1954	Effect / when to change
minClusters	10	Yield/quality trade-off. Lower to keep short backward recoils, at the price of poorly constrained fits.
gasDensity	6.616e-5	CATIMA density, H_2 at 600 torr / 293 K. Matches <code>media.geo H_600torr</code> exactly. The 2026-08-25 rebuild used 3.308e-5 (300 torr), half the material; it put the elastic peak at +0.2–0.3 MeV. Superseded 2026-09-25.
bFieldSign	–1	Experimental handedness.
refTrack	false	Do not re-attempt the unscented reference-track port; the UKF cannot match genfit’s analytic-Jacobian transport.
SetEnableEnergyStraggling	false	In the macro body.
SetEnableMultipleScattering	—	In the macro body; worth a test for the lowest-energy protons.
SetForceVertexOnBeamAxis	—	Vertex-constraint knob in <code>AtFitterUKF</code> (with <code>SetVertexZBias</code>), not exposed as a macro argument.

The UKF stores KE at two reference points, which is the cheapest available handle on energy-loss systematics: `GetKinematics()` is the vertex value (back-extrapolated to the beam axis with the lost energy added back), `GetKinematicsXtr()` is the value at the first cluster. On a1954 the difference is a median of only 0.082 MeV, moving the g.s. centroid by +0.055 MeV.

10.3 GENFIT knobs

Knob	Recommended	Effect / when to change
matEffects	kTRUE	The important one. False $\Rightarrow$ constant-momentum helix $\Rightarrow$ KE 30 % low below 6 MeV.
geoName	ATTPC_H600torr	Material effects are only as good as the density.
measSigma	4.0 mm	See Section 10.
minIter/maxIter	2 / 5	More iterations for badly seeded tracks.
thetaMin	10 (<i>pp</i>) / 5 (<i>pd</i>)	Rejects near-beam tracks; too high kills forward deuterons.
backwardSeedFix	kFALSE	Alternative seeding for backward tracks; untested here, worth a scan if backward-angle behaviour still looks odd.
pidGate	""	In-fitter Spyral PID gate; redundant when <code>gate_events_Be12.C</code> already gated.
SetELossFile	(unset)	Loading a proton dE/dx table would close the residual +16 % deficit below 2 MeV.

10.4 Drift velocity — the systematic most likely to be limiting you

The a1954 par file carries `DriftVelocity` 1.00 cm/ μs for ^{12}Be . In the sibling ^{14}C analysis of the *same experiment*, a scan showed the nominal value (1.30 there) was clearly too low, and moving to 1.45 improved the resolution by 18 % (FWHM 0.535 $\rightarrow$ 0.441 MeV) while pulling the g.s. from –0.36 to –0.09 MeV. Three criteria gave three slightly different optima — slope $\rightarrow$ 0 at 1.37, $\mu \rightarrow$ 0 at 1.49, best FWHM at 1.45 — so quote which one you used.

The same scan has not been done for ^{12}Be and is the highest-value next test. The recipe:

1. Copy `parameters/ATTPC.a1954_Be12.par` to `ATTPC.a1954_Be12_dv110.par` etc., changing only `DriftVelocity`.
2. Re-run *reco* for one run per point (drift velocity enters at the PSA z -calibration, so refitting alone is not enough), into per-point directories.
3. Gate and UKF-fit each point, then build an E_x cache per point with a distinct `outTag` (remember the leading underscore).
4. Tabulate g.s. μ , FWHM and the $dE_x/d\theta_{\text{cm}}$ slope against v_d , and pick your criterion.

10.5 Other levers on the resolution

- **Beam energy.** $dE_x/dE_{\text{beam}} \approx 0.01$ MeV per MeV, so zeroing the elastic peak constrains E_{beam} only to ± 20 MeV. Use the *shape* of the KE-vs- θ_{lab} locus (the explorers overlay the kinematic line) or an independent measurement.
- **Vertex- z dependence.** Resolution is flat in vertex z (a good sign for the calibration) but degrades for far vertices; a `vz < 600` cut gave FWHM 0.43 MeV but cost the weak 2^+ statistics.
- **E_x -vs- θ drift.** If the elastic centroid drifts with angle, that is a drift-velocity or KE-scale signature, not a statistical effect. The ^{14}C workspace has `pp/hyp_C14.C`, which discriminates the three hypotheses (wrong E_{beam} / wrong KE scale / empirical correction) by where it places known levels; porting it to ^{12}Be is straightforward.
- **Gates.** Redraw the PID polygon with `arclen > 200`; check that the deuteron gate is not leaking tritons (this happened in a1975 — genfit fits them with a perfectly good χ^2).

11 Known pitfalls

1. **Beam energy is not calibrated.** 155 MeV comes from zeroing the elastic peak, which as shown above barely constrains it. Every absolute E_x rides on it. Find the logbook value before quoting numbers. At the correct 600 torr the elastic peak sits at +0.003 (UKF) / −0.010 (GENFIT) MeV with 155 MeV — consistent, but in-sample: 155 was scanned against $6.5\text{e-}5$ g/cm³, which is essentially this gas.
2. **The gas is 600 torr, not 300.** A 2026-08-25 rebuild assumed 300 torr; it was wrong and is superseded (2026-09-25). The macro defaults (`gasDensity`, `geoName`) still carry the 300-torr values, so the drivers (`fitpipe_*Be12.sh`) pass `GASDENS=6.616e-5` and `GEONAME=ATTPC_H600torr` explicitly. The a1954 ^{14}C setting *was* 300 torr.
3. **Beam energy loss along v_z .** The (p,p') elastic peak rises with vertex z by 0.19 (UKF) / 0.16 (GENFIT) MeV/m: the beam enters at low v_z . CATIMA gives 9.1 MeV/m for ^{12}Be at 155 MeV in 600-torr H₂, which removes only $\sim 60\%$ of that drift; the explorer's $E_{\text{beam}}(v_z)$ control defaults to 9.1 (off).
4. **F: is not auto-mounted** after a reboot: `sudo mount -t drvfs F: /mnt/f`. Only the raw data and the big reco files live there; all fit outputs are local and survive.
5. **df / lies inside WSL.** It shows the guest view of a sparse VHDX. When the Windows C: drive fills, jobs get killed mid-write while `df` still reports free space. Watch `df /mnt/c` and compact the VHDX to reclaim.
6. **Kill ROOT jobs at 80 % memory** rather than waiting through swap.
7. **Two runs are timestamp-desynced** (0142, 0212) — exclude them (Section 5.2).

8. **IC amplitude is the positive overshoot**, not $-\text{min}$, and the gate is [500, 900], not the older [625, 750].
9. **θ conventions differ** between the gate stage (Spyral polar) and the fit output ($\approx 180^\circ$ minus that).
10. **outTag needs its leading underscore** — the cache name is built as "proton_kin" + outTag.
11. **ROOT Eve / 3D viewers crash under WSL**. Use the browser explorer or static PNGs. (Geant4 Qt/OpenGL, by contrast, does work under WSLg.)
12. **The GENFIT fix propagates**. AtGenfitter commit 5f87e625 affects a1975 and PUMA too; anything fitted with `matEffects = kTRUE` before it had the same silent $\text{ndf} \leq 0$ truncation and will change on a re-run.

12 Command cheat-sheet

```
# --- environment -----
export REPO=/home/yassid/fair_install/ATTPCROOTv2-OpenKF
export HERE=$REPO/macro/Unpack_HDF5/a1954_Be12/UKF
source /home/yassid/fair_install/FairSoft/install/bin/thisroot.sh
source $REPO/build/config.sh
export ROOT_INCLUDE_PATH=$REPO/build/include:/home/yassid/fair_install/FairRootInstall/include
sudo mount -t drvfs F: /mnt/f          # if F: is not mounted
root -b -q $REPO/geometry/ATTPC_H600torr.C      # once per clone

# --- stage 1-2: reco + auxiliary streams -----
cd $HERE
./reco_hdb_F.sh      "run_0143 run_0144" 2      # -> /mnt/f/a1954_Be12_reco_hdb/
./frib_F_batch.sh    "run_0143 run_0144" 4      # -> ~/a1954_Be12_reco_hdb_slim/*_FRIB.root
./slim_cache_batch.sh "run_0143 run_0144" 4      # -> ~/a1954_Be12_reco_hdb_slim/*_slim.root

# --- stage 3: checks and gates -----
root -b -q -l "$HERE/pid/tsync_check.C(\"run_0143\")"
root -b -q -l "$HERE/pid/ic_Be12.C(\"run_0143\")"
root -l      "$HERE/pid/draw_gate_Be12.C(\"proton_12Be\")"      # interactive

# --- stage 4-5: gate + fit -----
./fitpipe_Be12.sh    "run_0143 run_0144" 4      # (p,p'): gate -> UKF -> GENFIT
./fitpipe_pd_Be12.sh "run_0143 run_0144" 4 0    # (p,d) : gate -> UKF(deuteron)
./genfit_pd_Be12.sh  "run_0143 run_0144" 4      # (p,d) : GENFIT

# --- stage 6: physics -----
root -b -q -l "$HERE/pp/ex_Be12.C(\"run_0143,run_0144\", \"~/a1954_Be12_genfix/\", 155, 5, \"_pp_genfit\", 1.007825, 12.026921, \"12Be(p,p')\", \"genfit\")"
$HERE/pp/open_explorer.sh pp
```

Output directories

/mnt/f/a1954_Be12_reco_hdb/	full reco (big)
~/a1954_Be12_reco_hdb_slim/	slim AtPatternEvent cache + FRIB
~/a1954_Be12_fit/in/	gated fit inputs (reusable)
~/a1954_Be12_fit/	(p, p') UKF + pre-fix GENFIT
~/a1954_Be12_genfix/	(p, p') fixed GENFIT
~/a1954_Be12_fit_pd/	(p, d) UKF
~/a1954_Be12_genfix_pd/	(p, d) fixed GENFIT
~/a1954_Be12_sigtest/	sigma-swap covariance test
\$HERE/pp/plots/	E_x caches and PNGs

Further reading in the workspace

README.md	the terse version of this guide
ANALYSIS_REPORT.md	the first end-to-end physics pass
pp/FITTER_COVARIANCE_TEST.md	the full UKF-vs-GENFIT investigation
pid/PID_COMPARISON.md	Spyral vs ATTPCROOT PID
